CALCULATION-02

SQUARE (वर्ग)

CLASS NOTES

Duplex Method of calculating Squares

Duplex of a single digit number (a) is its square.

$$D(a) = a^2$$

Examples:

D (4) = 16

D (7) = 49

D (2) = 4

D (8) = 64

D (9) = 81

Duplex Method of calculating Squares

Duplex of a two digit number (ab) is twice of their product.

D(ab) = 2ab

Examples:

D (10) = 0

D (12) = 4

D (93) = 54

D (51) = 10

D(13) = 2x1x3 = 6

D(23) = 2x2x3 = 12

D(46) = 2x4x6 = 48

D(84) = 2x8x4 = 112

D(94) = 2x9x4 = 72

Duplex Method of calculating Squares

Duplex of a three digit number (abc) is twice the product of the outer pair + the square of the middle digit

$$D(abc) = 2ac + b^2$$

$D(134) = 2 \times 1 \times 4 + 3^2 = 17$

$D(512) = 2 \times 5 \times 2 + 1^2 = 21$

$D(234) = 2 \times 2 \times 4 + 3^2 = 34$ ✓

Examples:

D (123) = 10

D (512) = 21

D (108) = 16

D (287) = 92

Duplex Method of calculating Squares

Duplex of a four digit number (abcd) is twice the product of the outer pair + twice the product of the inner pair

D(abcd) = 2ad + 2bc

$= 2(ad + bc)$

$D(1234) = 2(1 \times 4 + 2 \times 3)$

$= 2(4 + 6)$

$= 20$

Examples:

D (1234) = 20

D (5079) = 90

D (3841) = 16

D (287) = 92

$$D(2345) = 2(2\times5+3\times4)$$
$$= 2(10+12)$$
$$= 44$$

$$D(4865) = 2(35+48)$$
$$= 2\times83$$
$$= \underline{166}$$

Square of Two-Digit Numbers

$(ab)^2$ = D(a) / D(ab) / D (b)

Example

$(23)^2$ = D(2) / D(23) / D (3)

= 4 / 12 / 9

= 529

$(23)^2 = D(2) / D(23) / D(3)$

$= 4 / 12 / 9$

$= 529$

$(86)^2 = D(8) / D(86) / D(6)$

$= 64 / 96 / 36$

$= 7396$

$(23)^2 =$ 1 (carry) | 4 | 2 | 9

$= 529$

$(73)^2$ = 49 / 42 / 9

= 5329 ✓

$(53)^2$ =

$(77)^2$ = 49 98 49

5929

$(43)^2$ = 16 / 24 / 9

1849

$(79)^2 = 49 / 126 / 81$

$= 6241$

$(88)^2 = 64 / 128 / 64$

$= 7744$

Square of Three-Digit Numbers

$(abc)^2$ = D(a) / D(ab) / D (abc) / D (bc) / D (c)

Example

$(123)^2$ = D(1) / D(12) / D (123) / D (23) / D (3)

= 1 / 4 / 10 / 12 / 9

= 15129

$$(abc)^2 = D(a) / D(ab) / D(abc) / D(bc) / D(c)$$

$$\star\ (123)^2 = D(1) / D(12) / D(123) / D(23) / D(3)$$

$$= 1 / 4 / 10 / 12 / 9$$

$$= \underline{1\,5\,12\,9}$$

$\star$ $(234)^2 = D(2)/D(23)/D(234)/D(34)/D(4)$

$= 4/12/25/24/16$

$=$

$= 54756$

$\star$ $(984)^2 = 81/144/190/112/49$

$= 974169$

$(217)^2$ =

4/4/29/14/49

= 47089

~~$(108)^2$ =~~

$(156)^2$ =

1/10/37/60/36

= 24336

$(283)^2$ =

4/32/76/48/9

= 80089

$(514)^2$ =

= 25/10/41/8/16

264196

$(789)^2$ =

= 49/112/190/144/81

622521

Square of Four-Digit Numbers

$(abcd)^2$ = D(a) / D(ab) / D (abc) / D (abcd) / D (bcd) / D (cd) / D (d)

Example

$(1234)^2$ = D(1) / D(12) / D (123) / D (1234) / D (234) / D (34) / D (4)

= 1 / 4 / 10/ 20 /25 24 / 16

= 1522756

$$(abcd)^2 = D(a)/D(ab)/D(abc)/D(abcd)/D(bcd)/D(cd)/D(d)$$

$$(1234)^2 = D(1)/D(12)/D(123)/D(1234)/D(234)/D(34)/D(4)$$

$$= 1/4/10/20/25/24/16$$

$$= 152256$$

$(\overline{3456})^2 =$ 9/24/46/76/73/60/36

$=$ 11943936

$(7591)^2 =$ 49/70/151/104/91/18/1

$=$ 57623281 ✓

$(2023)^2$ =

$(5841)^2$ =

Mixed Practice

$(46)^2$ =

$(79)^2$ =

$(203)^2$ =

$(341)^2$ =

9/84/22/8/1

116281

Mixed Practice

$(412)^2$ =

$(384)^2$ =

$(759)^2$ =

$(983)^2$ =

Special Case - 01

$(1008)^2 = (1008-8)(1008+8)+8^2$

$= 1000 \times 1016 + 64$

$= 1016064$

Square of a number which is nearer to 10^x

$(98)^2 = (98+2)(98-2)+2^2$

$= 100 \times 96 + 4 = 9604$

$(103)^2 = (103-3)(103+3)+3^2$

$= 100 \times 106 + 9 = 10609$

$(993)^2 = (993+7)(993-7)+7^2$

$= 1000 \times 986 + 49 = 986049$

$(1008)^2 =$

Special Case-02

Let's observe

$3^2 = 0\ \ 9$

$33^2 = 1089$

$333^2 = 110889$

$3333^2 = 11108889$

$6^2 = 3\ \ 6$

$66^2 = 4356$

$666^2 = 443556$

$6666^2 = 44435556$

$9^2 = 8\ \ 1$

$99^2 = 9801$

$999^2 = 998001$

$9999^2 = 99980001$

$6^2 = 36$

$66^2 = 4356$

$666^2 = 443556$ ✓

$6666^2 = 44435556$

$3^2 = 09$

$33^2 = 1089$

$333^2 = 110889$

$3333^2 = 11108889$

$9^2 = 81$

$99^2 = 9801$

$999^2 = 998001$

$9999^2 = 99980001$

Special Case-03

Let's observe

$1^2 = 1$

$11^2 = 121$

$111^2 = 12321$

$1111^2 = 1234321$